

SANJAY S

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RESEARCH INTERESTS

- Superconducting Quantum Hardware (Transmon, Fluxonium)
- Quantum Processor Architecture and Scalable Qubit Systems
- Microwave Resonators and Circuit Quantum Electrodynamics
- Quantum Control Electronics and RF Signal Systems

EDUCATION

Chennai Institute of Technology and Applied Research, Chennai

Aug 2023 – Present

B.E. Electronics and Communication Engineering | CGPA: 9.15

Cholan Matriculation Higher Secondary School, Kanchipuram

Jun 2021 – May 2022

12th Grade | Percentage: 92%

Cholan Matriculation Higher Secondary School, Kanchipuram

Jun 2019 – Apr 2020

10th Grade | Percentage: 98%

EXECUTIVE SUMMARY

- Electronics and Communication Engineering student focused on superconducting quantum hardware, working on the design and electromagnetic analysis of transmon qubits and microwave resonators using Qiskit Metal, scQubits, KQCCircuits, Qutip, Ansys HFSS, Ansys Q3D, Keysight ADS and QuantumPro, with experience in quantum processor architecture through a flip chip superconducting transmon processor project selected as a finalist in the VLSID Design Contest.
- Currently gaining practical exposure to experimental quantum systems at C-DAC Bengaluru, learning quantum control electronics through IP development for RF ADC, RF DAC, and RF data converter modules, and working with laboratory instruments including DSO, lasers, RF drivers, RF attenuators, optical couplers, polarization beam splitters, and photon detection setups used in photonic quantum experiments.
- Built a machine learning-based randomness source attribution framework during the SparQ Internship Program at QNu Labs, and separately developed a dispersive shift (χ) optimization tool for superconducting qubit and resonator design as part of Quant-A-Thon 2026.

WORK EXPERIENCE

Research Intern | Quantum Technology Group, C-DAC Bengaluru

Jan 2026 – Aug 2026

- Designed and simulated superconducting qubits using Qiskit Metal, scQubits, KQCCircuits, Ansys HFSS, Ansys Q3D, and Keysight ADS QuantumPro for quantum processor development.
- Developed IP modules for RF ADC, RF DAC, and RF data converters while studying quantum control electronics and signal generation techniques such as NCO and AWG.
- Worked with quantum photonic laboratory systems including DPS-QKD setups and optical instruments such as lasers, optical couplers, polarization beam splitters, photon detectors, and digital storage oscilloscopes; performed photonic coincidence experiments.

Project Intern, RNG Randomness Quality Detection (ML) | SparQ Internship Program, QNu Labs Private Limited

May 2026 – Jul 2026

- Contributed to RSAF, a hierarchical, confidence-oriented machine learning framework that attributes binary randomness streams to algorithmic (PRNG) or physical (TRNG/QRNG) sources, evaluated on 37 binary datasets (~4.15 GB) including QNu Labs' proprietary photonic QRNG output.
- Trained and tuned the five-model tree-based ensemble (CatBoost, XGBoost, LightGBM, Random Forest, Extra Trees) used to classify a 64-dimensional statistical feature set spanning entropy, spectral, correlation, and complexity descriptors, and benchmarked model performance across accuracy, precision, recall, and F1-score.
- Framework achieved a mean generator-level attribution confidence of 81.21%, a calibrated Brier score of 0.16 (down from 0.21 pre-calibration), and an ensemble F1-score of 73.2%, outperforming every individual base learner.

PROJECTS

RSAF: A Hierarchical, Confidence-Oriented Framework for Randomness Source Attribution | SparQ Internship, QNu Labs *May 2026 – Jul 2026*

- Built a 64-feature statistical fingerprinting pipeline (entropy, spectral, correlation, and complexity descriptors) and a five-model calibrated ensemble to distinguish algorithmic (PRNG) from physical (TRNG/QRNG) randomness sources.
- Applied isotonic probability calibration and hierarchical window-to-file-to-generator voting to convert raw classifier scores into a stable, calibrated attribution confidence score.

- Supported SHAP-based explainability and the team's Randomness Quality Index (RQI), an independent 0–100 composite metric for entropy source quality, used alongside the attribution decision.

Dispersive Shift (χ) Optimizer for Superconducting Qubit Design

RIT Quant-A-Thon 2026 – Finalist

- Built a closed-form inverse-design engine that converts a target qubit frequency, resonator frequency, and dispersive shift (f_q, f_r, χ) into a complete, fabricable transmon and readout-resonator geometry.
- Implemented a multi-fidelity Bayesian optimization framework (Optuna TPE) across an analytic $\rightarrow$ Q3D/MoM $\rightarrow$ HFSS+EPR fidelity ladder, reserving full electromagnetic solves for verification only to keep the design loop fast.
- Developed the Streamlit-based design GUI and forward-model cross-checks against Keysight ADS QuantumPro, validating computed transmon parameters to within 0.04–0.6% of published and measured reference values.
- Represented Team Transmon Titans as a finalist in the RIT Quant-A-Thon 2026 dispersive-shift optimization track.

4 Qubit Flip Chip Transmon for Behavioral Fraud Detection

VLSID Design Contest 2026 – Finalist

- Designed superconducting transmon qubit and resonator structures using Qiskit Metal with 3D flip chip architecture to reduce routing congestion and electromagnetic crosstalk.
- Performed electromagnetic analysis of qubit coupling and coherence, with simulation results indicating improvement from $\sim 3.02 \mu\text{s}$ to $\sim 4.94 \mu\text{s}$ and crosstalk reduction from $\sim -37 \text{ dB}$ to $\sim -55 \text{ dB}$.
- Developed a behavioral biometric authentication model using quantum machine learning concepts for human-bot classification.

FPGA-Based Quantum Control Electronics: RF Data Converter IP & RTL TRNG

C-DAC Bengaluru – 2026

- Implemented Zynq UltraScale+ RF Data Converter block design with AXI SmartConnect.
- Implemented complete TRNG in Verilog: dual ring-oscillator entropy source, Von Neumann + LFSR conditioning, XOR sampler, repetition test, adaptive health monitor, fault FSM, and full top-level testbench integration.

Design of a Single Transmon Qubit with an Integrated Readout Resonator

Sep 2025

- Designed and analyzed a single transmon qubit coupled to a readout resonator using Qiskit Metal and Ansys HFSS & Ansys Q3D.
- Performed geometric optimization, EM analysis and coupling-strength characterization for qubit–resonator interaction.
- Achieved qubit frequency of 5.51 GHz and anharmonicity of -377 MHz , meeting all design targets.
- Validated key parameters: $E_J/E_C = 41.2$, $g = 119 \text{ MHz}$, $\chi = 3.89 \text{ MHz}$, and charge dispersion = 62 kHz.
- Obtained a simulated T_1 of 46.1 μs , demonstrating strong coherence and suitability for practical quantum computing.

TECHNICAL SKILLS

- Programming: Python, C, Verilog
- Quantum Software: Qiskit, Cirq, QuTiP
- Quantum Hardware Design: Qiskit Metal, scQubits, QCCircuits
- Electromagnetic and RF Simulation: Ansys HFSS, Ansys Q3D, Keysight ADS, QuantumPro
- Hardware Design and Control Electronics: Vivado, RTL Design, FPGA IP development for RF ADC, RF DAC, RF Data Converters
- Operating Systems: Linux, Windows
- Soft Skills: Interdisciplinary teamwork across institutes, clear technical communication, project coordination

WORKSHOPS

Quantum Superconducting Circuit Design Program – Womanium Quantum, Quantum Technology / Semiconductor Device / Nanotechnology Simulation and Experimental Characterization – Impulse Technology, Qiskit Fall Fest 2024 – IBM Quantum, Demystifying Quantum Computing – IAPT, Short Term Training Program on Quantum Computing

SEMINARS

Novel Superconducting Qubits – Google Quantum AI, Qiskit Global Summer School 2024 – IBM Quantum, Quantum Artificial Intelligence for Renewable and Sustainable Energy

COURSES

The Hardware of a Quantum Computer – TU Delft, Fundamentals of Superconducting Qubit – QuTech Academy, Quantum Computing with Semiconductor Technology Professional Certificate – TU Delft, Introductory Course on Quantum Computing – IISER Pune, Qubit by Qubit's Introduction to Quantum Computing – The Coding School, Introduction to Quantum Computing: Quantum Algorithms and Qiskit – IIT Madras

ACHIEVEMENTS

- Winner, MSME Idea Hackathon 4.0 – ₹15 Lakhs award for neuromorphic EO payload small satellite concept.
- Finalist, VLSID Design Contest 2026 – 3D flip chip 4 qubit superconducting transmon processor design.
- Finalist, RIT Quant-A-Thon 2026 – Dispersive shift (χ) optimization for superconducting qubit design.
- Finalist, Amaravati Quantum Valley Hackathon – Quantum machine learning solution for fraud detection.